

CL-2 Assignment-1

Report

Divyanshu Giri
2024114009

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What are the shortest words in dataset? Comment on these words.

Function words are usually the shortest words in any dataset and similarly we can see that the shortest words in this dataset. There are some short words accounted in the dataset which usually are numerical figures and abbreviations. Commonly used short verbs(work, copy, fix, run, etc.) are also there but are present in smaller percentage.

Write a short note on: “Are word lengths optimized for efficient communication?”

Please connect your answer to the following research paper:
Word lengths are optimized for efficient communication by Steven T. Piantadosi, Harry Tily, and Edward Gibson.

According to Zipf, “the magnitude of words tends, on the whole, to stand in an inverse (not necessarily proportionate) relationship to the number of occurrences”. Zipf’s explanation implied that the length of a word is largely dependent on frequency, enabling speakers to save effort through shortened most frequent words. Information can be conveyed as concisely as possible by giving the most frequently used meanings the shortest word forms, much like in variable-length (e.g., Huffman) codes.

This new research paper refutes this sole claim by suggesting that : the amount of information conveyed by a word should be linearly related to the amount of time it takes to produce— approximately, its length—to convey the same amount of information in each unit of time and not the frequency. Applying big corpora for 11 languages, they determined that the length of a word is associated more closely with the informativeness of a word in context than with frequency in isolation. More predictable words (and therefore less information-bearing) are shorter, and less predictable words are longer. This guarantees that the information transfer rate is fairly consistent. To formalize this, we can

consider two random variables, C for contexts and W for words, with a joint distribution $P(C, W)$ given by the natural statistics of language use. The average amount of information conveyed by a particular word w is given by the following:

$$-\sum_c P(C = c \mid W = w) \log P(W = w \mid C = c).$$

Empirically, they found that correlations between information content and word length were consistently stronger than those between frequency and length, even when controlling for each factor. In many cases, the partial correlation between frequency and word length vanished once information content was accounted for, whereas the reverse was not true. This indicates that frequency effects are largely explained by information content.

The data when represented in graph illustrated that length is more predictable from information content than from frequency. Frequency does not predict word length substantially for lower-frequency words: words in low unigram probability bins have approximately the same average length. In contrast, length varies as a function of information over almost its entire range of values, with the exception of just the few words with the very lowest information content.

In summary, this research implies that word lengths are optimized for effective communication, not just by frequency-based compression, but by mapping length to contextual informativeness. This redescribes linguistic economy: languages are not only optimized to minimize effort, but to ensure a stable and efficient information flow over different contexts.